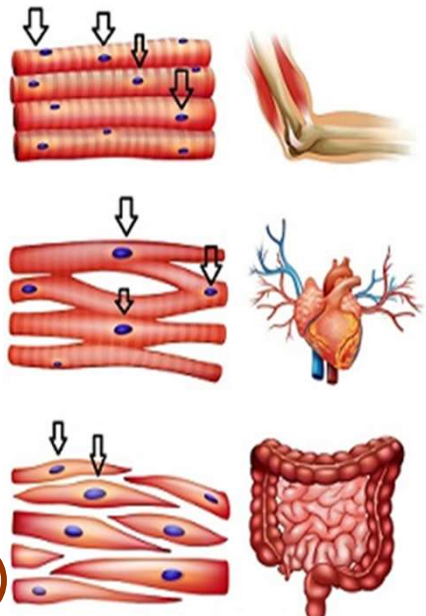


بِسْمِ اللَّهِ
الرَّحْمَنِ الرَّحِيمِ

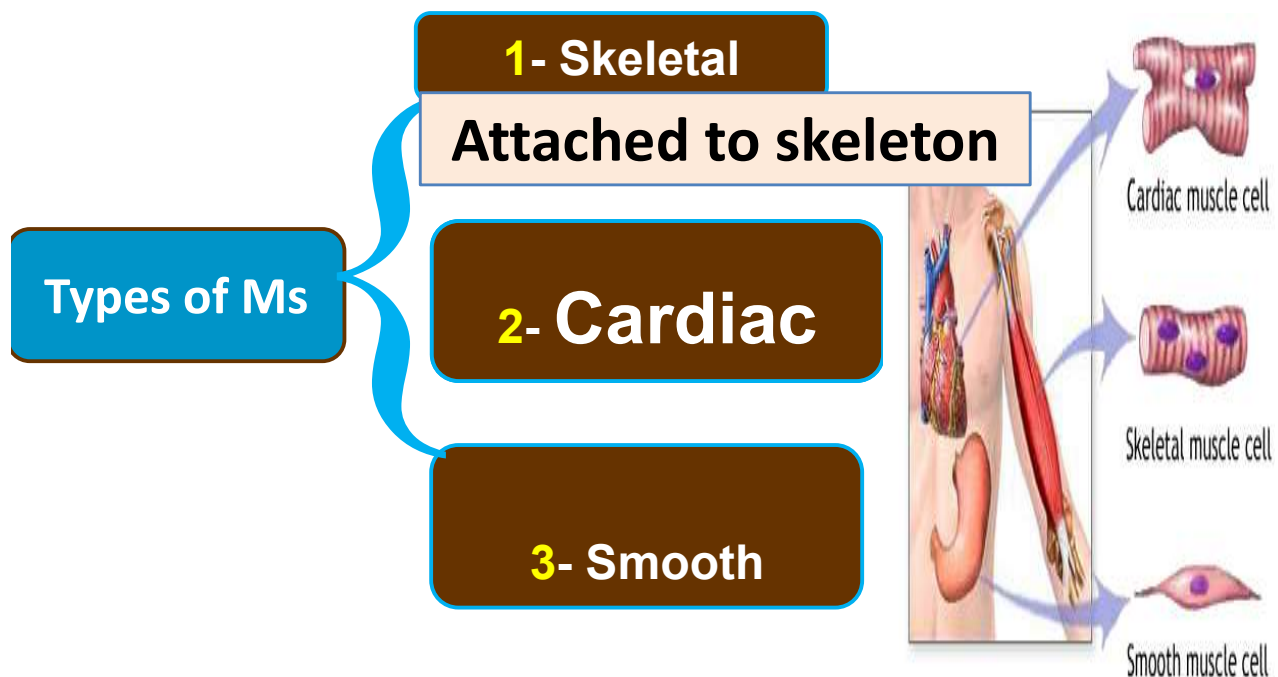
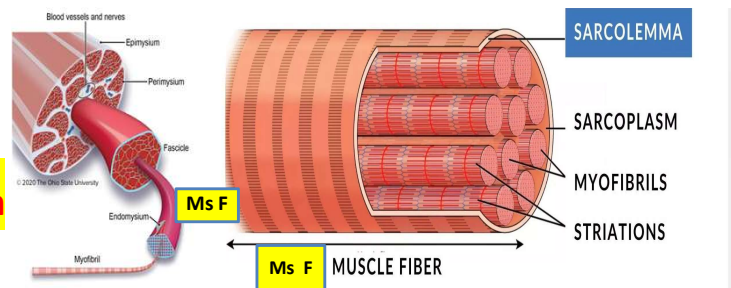
Muscular Tissue

Prof . Eman Abas

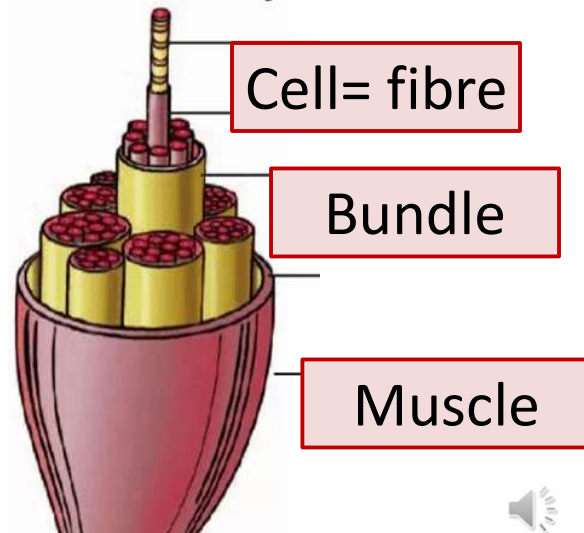
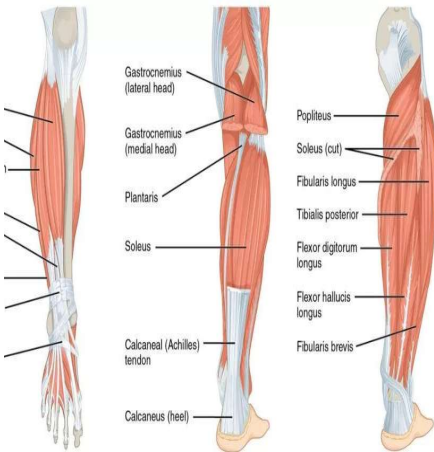


General characters

- Origin: **mesoderm**
- Formed of cells=**Ms Fibre.**
- Contain **myofibrils** have **actin& myosin** →
interact → Ms contraction
- CM = **sarcolemma**
- Cytop.= **sarcoplasm**
- sER=**sarcoplasmic reticulum**



Skeletal muscle



CT covering of SK MS

CT is arranged as

1-**Epimysium**=

Around **whole muscle**

2-**Perimysium**=

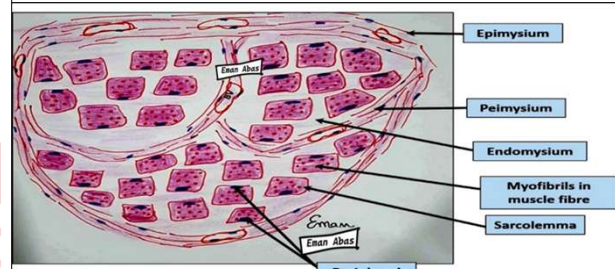
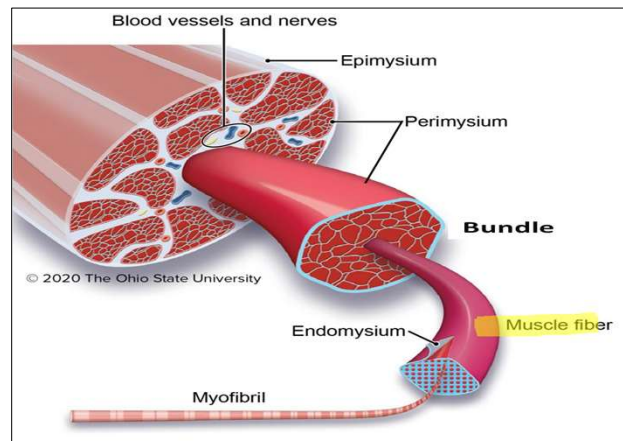
Around **bundles**

3-**Endomysium**=RF

Between **fibres**

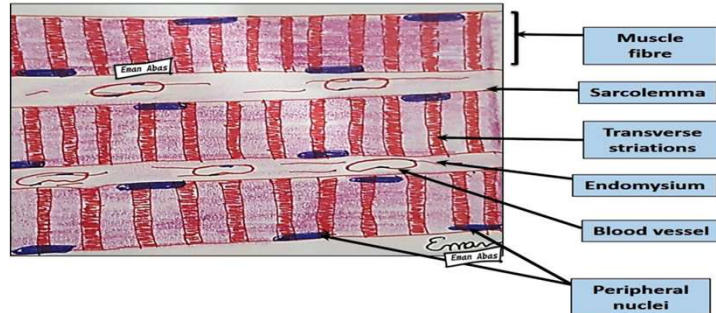
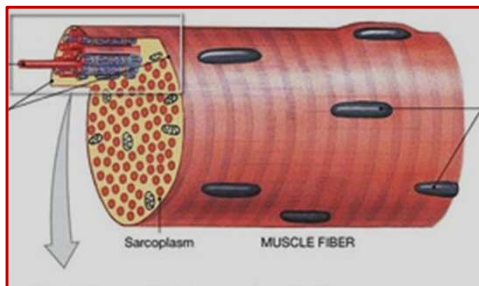
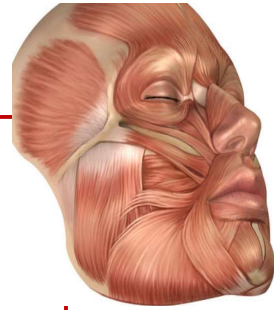
Fn of CT=

1. **Connect Ms F: force transmission**
2. **Carry BL V& N: nutrition**



LM of skeletal muscle fibre

- SHAPE:** cylindrical **non branched** (- face& tong)
- Nuclei:** multiple – oval - peripheral
- Sarcoplasm:** deep acidophilic+ TS striation (**in LS**)



EM of skeletal muscle fibre

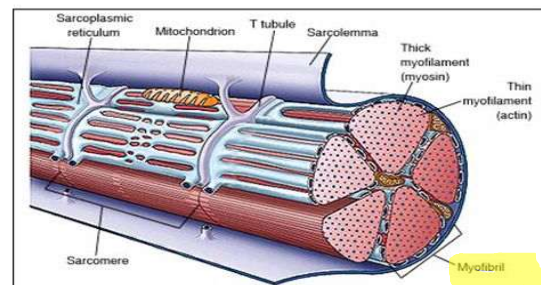
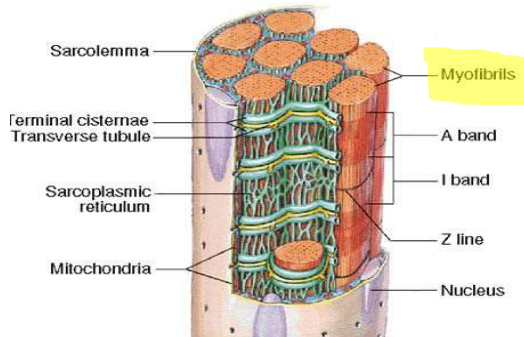
1- **Nucleus?** EUchromatic

2- **Organelles**

3- **Inclusions**

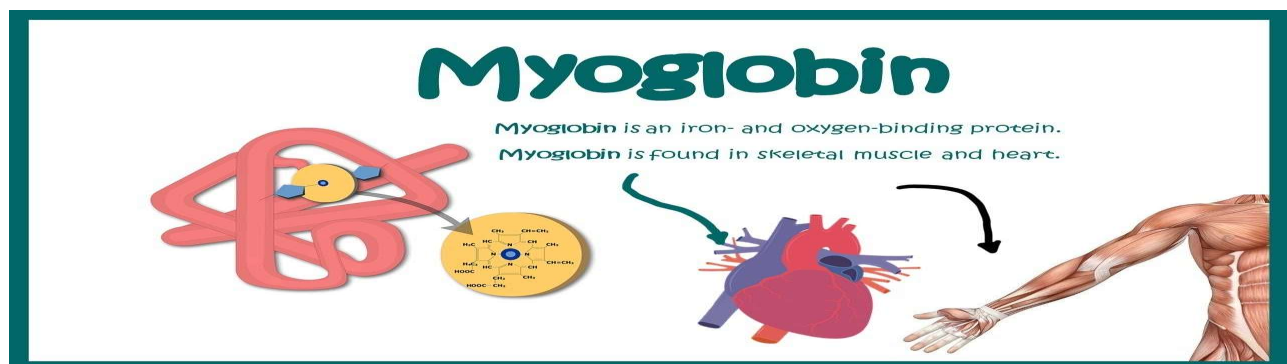
2- Organelles

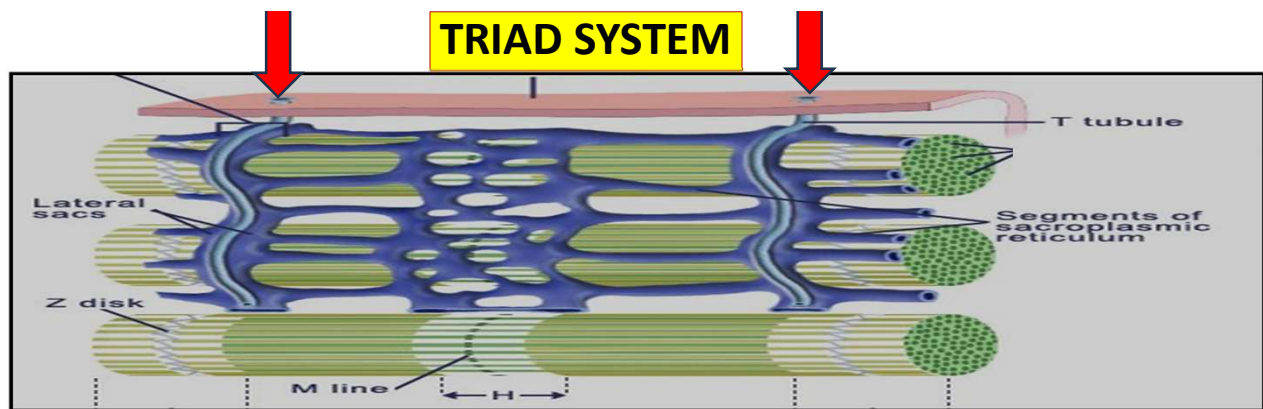
- **Myofibrils** : many –cylindrical- //long axis of ms?? **Contraction**
- **SR** : =SER- form **network** around **Myofibrils**. fn: **Ca** store& release
- **Mitochondria** : many- **rows** between MF → **ENERGY**



3- Inclusions

- **Myoglobin** : O_2 binding pigmented. Ptn → O_2 ??????
- **Glycogen& lipid** : → Energy-



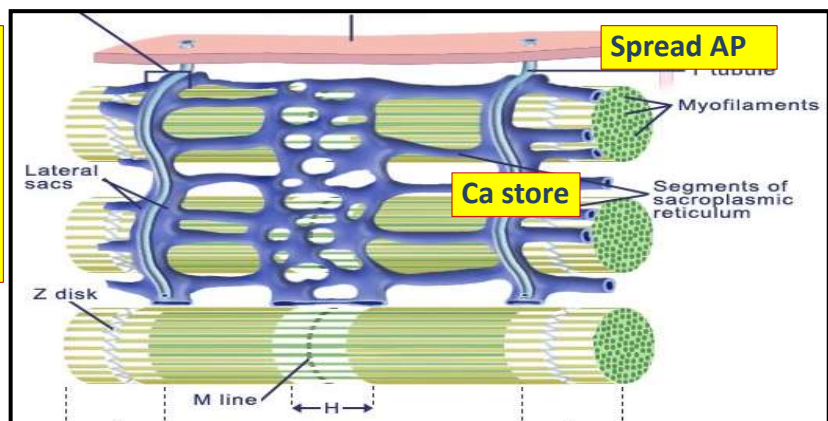


= **1 T tubule** = transverse tubule = sarcolemma invagination around MF

- continuous w exterior **Transmit nerve impulse [Action Potential]**

+ **2 SR** = 2 Sarcoplasmic reticulum or terminal cisternae **Ca store & release**

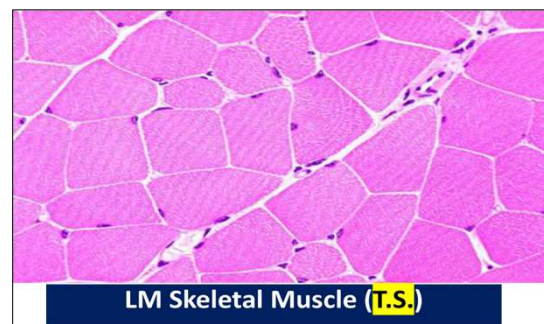
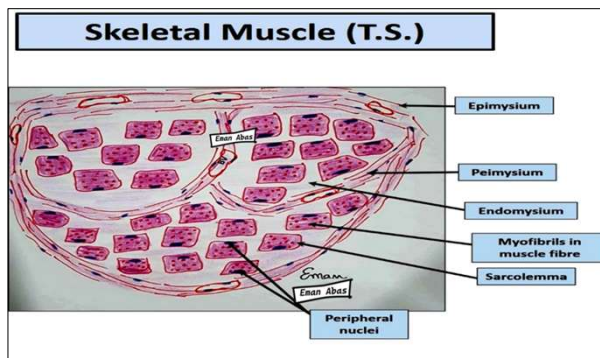
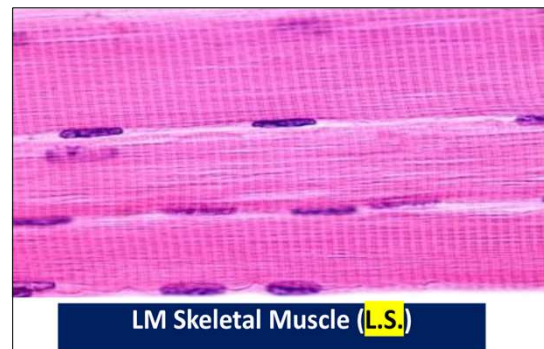
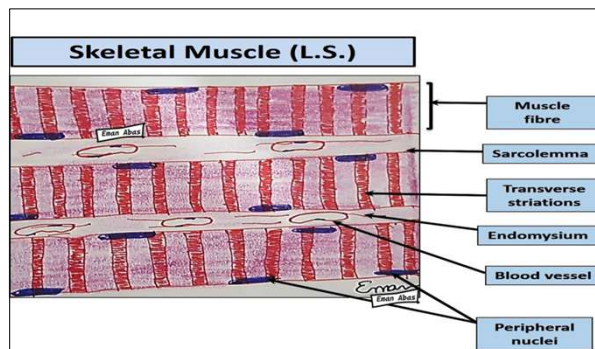
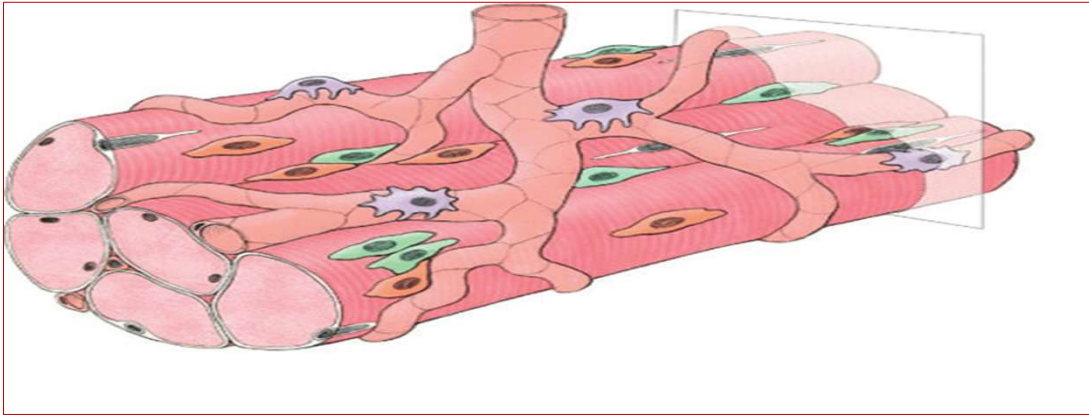
Role of triad system in muscle contraction.



Depolarization of sarcolemma by nerve impulse →
Spread into muscle through **T tubule** → **Ca⁺⁺** release
from **SER** → sliding of actin over myosin →
contraction.

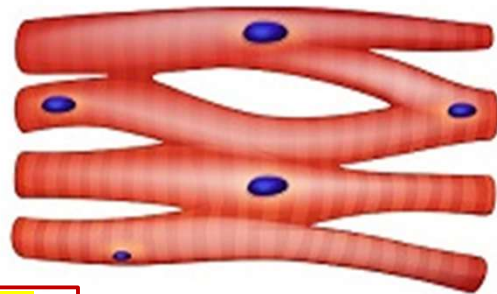
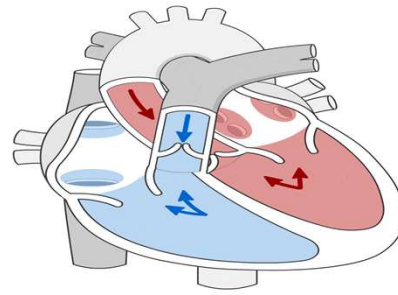
Regeneration of skeletal muscle

By satellite cells



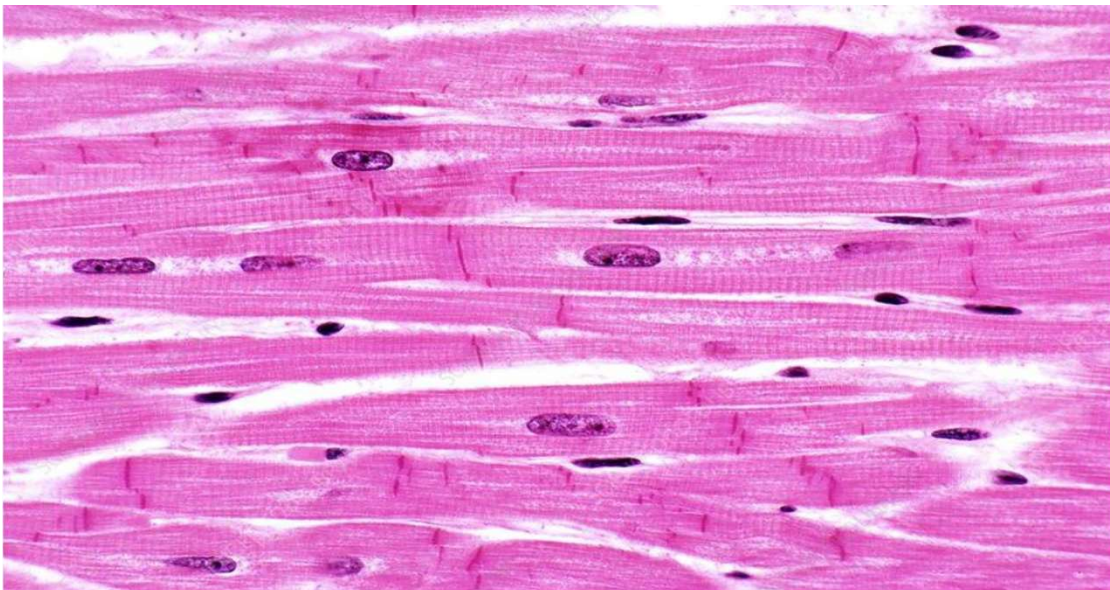
Cardiac muscle

- **Site:** wall of heart
- **Shape:** cylind. **Branched?????**
with **single central oval** nucleus
and **LESS clear TS?????**
- Has **diad not triad** system=??



Cannot regenerate: Heal by fibrosis?

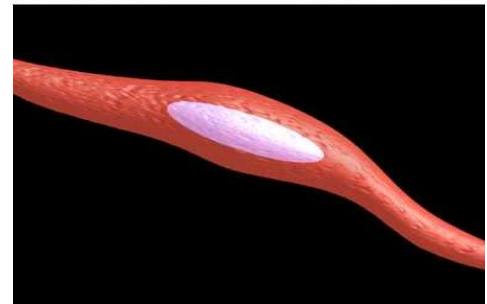
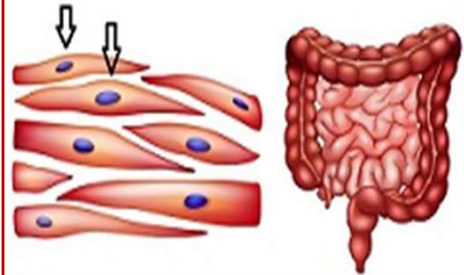
Cardiac Muscle



LM Cardiac Muscle

Smooth Muscle

- **Site:** wall of viscera & **blood vessels**
- **Shape:** spindle **non branched** with **single central oval** N and **NO TS**
- **NO** tubular system
- **Regenerate** from UMCs



Differences between m. fibers	Skeletal 	Cardiac 	Smooth 
Site	Skeleton	Heart	Viscera
Shape	Cylindrical	Cylindrical	spindle 
Size	Large	Medium	Small
Branching	Not Rare 	Branched	Not branch
Sarcolemma	Thick	Very thin	Thin
Striations	Clear	Non clear	Absent
Tubular system	Triad	Diad	Absent
regeneration	satellite	NO	Pericytes-UMCS
Nuclei (oval)	Multiple & peripheral	Single & central	Single & central
Action	Voluntary	Involuntary Autonomic	
Innervation	Motor		

Thank you
Eman Abas 